

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

1. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)

Program code:

```
% Wireless Healthcare Monitoring: Cloud-Edge Communication Model
```

```
clear; clc;
```

```
%% 1. Simulate Patient Data (Sensor Layer)
```

```
time_duration = 100; % Simulation time in seconds
```

```
fs = 10; % Sampling frequency (Hz)
```

```
t = linspace(0, time_duration, time_duration * fs);
```

```
% Simulate Heart Rate (Normal range ~60-90 bpm, with random spikes)
```

```
baseline_hr = 75 + 5 * randn(1, length(t));
```

```
anomaly_indices = randi([1, length(t)], 1, 5); % 5 random anomaly events
```

```
baseline_hr(anomaly_indices) = baseline_hr(anomaly_indices) + 50; % Add severe spikes
```

```
%% 2. Edge Processing Layer (Local Node / Wearable)
```

```
disp('--- Edge Node Processing ---');
```

```
threshold_high = 100; % bpm threshold for tachycardia
```

```
threshold_low = 50; % bpm threshold for bradycardia
```

```
edge_buffer = zeros(1, length(t));
```

```
transmission_flags = false(1, length(t)); % Flags data needing cloud upload
```

```
for i = 1:length(t)
```

```
    current_hr = baseline_hr(i);
```

```
    % Edge Decision Logic (Real-time, Low Latency)
```

```
    if current_hr > threshold_high || current_hr < threshold_low
```

```
        transmission_flags(i) = true; % Flag for cloud transmission
```

```
        % Edge can trigger immediate local alarm here
```

```
    end
```

```
end
```

```
%% 3. Cloud Processing Layer (Remote Server)
```

```
disp('--- Cloud Node Receiving ---');
```

```
cloud_database = []; % Stores only critical data
```

```
transmission_delay = 0.05; % Simulated network latency (seconds)
```

```
for i = 1:length(t)
```

```
    if transmission_flags(i)
```

```
        % Transmit to cloud (simulate network overhead)
```

```
        pause(transmission_delay);
```

```
        % Cloud stores event with timestamp
```

```
        cloud_database = [cloud_database; t(i), baseline_hr(i)];
```

```
    end
```

```
end
```

```
%% 4. Architecture Metrics (Evaluation)
```

```
total_data_points = length(t);
```

```

transmitted_points = size(cloud_database, 1);
bandwidth_saved = ((total_data_points - transmitted_points) / total_data_points) * 100;

fprintf('Total Sensor Readings: %d\n', total_data_points);
fprintf('Readings Sent to Cloud: %d\n', transmitted_points);
fprintf('Bandwidth Reduction at Edge: %.2f%%\n', bandwidth_saved);

% Plotting the Architecture Flow
figure('Name', 'Healthcare Cloud-Edge Architecture');
plot(t, baseline_hr, 'b', 'DisplayName', 'Raw Sensor Data (Edge)');
hold on;
plot(cloud_database(:,1), cloud_database(:,2), 'ro', 'MarkerSize', 8, 'DisplayName',
'Anomalies Sent to Cloud');
yline(threshold_high, 'r--', 'High Threshold');
xlabel('Time (s)');
ylabel('Heart Rate (bpm)');
title('Edge Data Filtering before Cloud Transmission');
legend;
grid on;

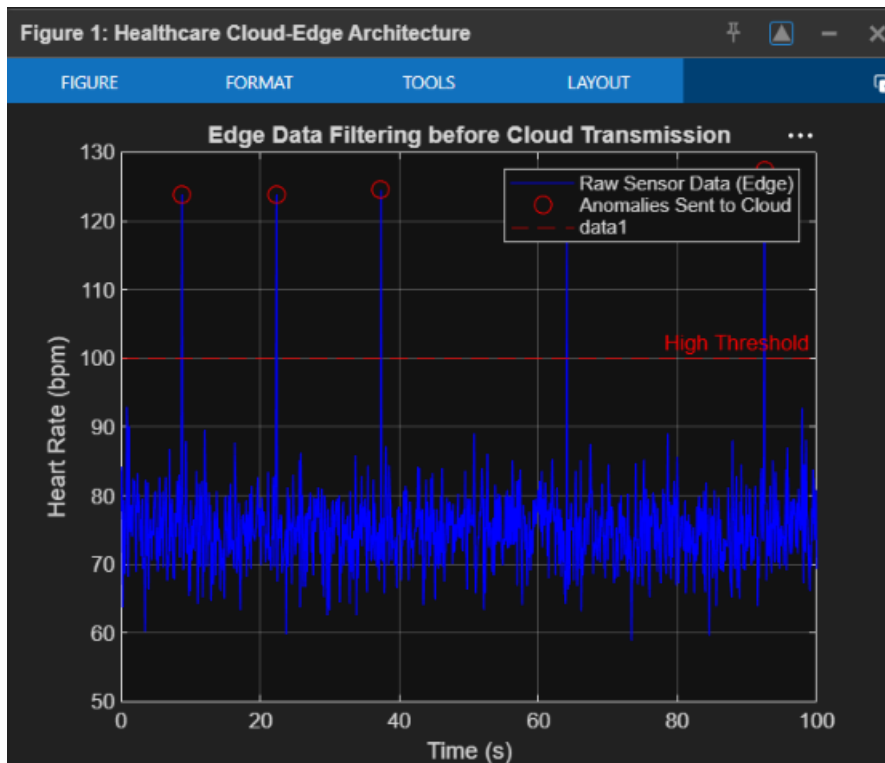
```

Output:

```

Command Window
--- Edge Node Processing ---
--- Cloud Node Receiving ---
Total Sensor Readings: 1000
Readings Sent to Cloud: 5
Bandwidth Reduction at Edge: 99.50%
>>

```



2. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)

Program code:

```
% Industrial Automation: Real-time Edge Data Processing
clear; clc;
```

```
%% 1. Plant Floor Data (High-Frequency Industrial Sensors)
```

```
sim_time = 50; % seconds
```

```
fs = 100; % High sampling rate for vibration analysis (100 Hz)
```

```
time = linspace(0, sim_time, sim_time * fs);
```

```
% Simulate Motor Vibration (Normal sine wave + random noise +
increasing wear)
```

```
motor_vibration = sin(2*pi*5*time) + 0.5*randn(1, length(time));
```

```
% Inject a mechanical fault halfway through
```

```
fault_idx = length(time)/2;
```

```
motor_vibration(fault_idx:end) = motor_vibration(fault_idx:end) +  
linspace(0, 3, length(time)-fault_idx+1);
```

```
%% 2. Edge Controller Architecture (Programmable Logic  
Controller / Edge Gateway)
```

```
disp('--- Edge Controller: Real-Time Protection Loop ---');
```

```
vibration_threshold = 3.5; % Safety limit
```

```
machine_status = 'RUNNING';
```

```
% Edge computes RMS (Root Mean Square) locally in small  
windows to avoid
```

```
% sending high-frequency data to the cloud.
```

```
window_size = 50;
```

```
num_windows = floor(length(time) / window_size);
```

```
cloud_aggregated_data = zeros(num_windows, 2); % [Timestamp,  
RMS_Value]
```

```
for w = 1:num_windows
```

```
    idx_start = (w-1)*window_size + 1;
```

```
    idx_end = w*window_size;
```

```
    current_window = motor_vibration(idx_start:idx_end);
```

```
    % Edge Processing: Calculate RMS
```

```
    rms_val = sqrt(mean(current_window.^2));
```

```
    timestamp = time(idx_end);
```

```
    % Edge Decision: Real-time actuator trigger
```

```
    if rms_val > vibration_threshold && strcmp(machine_status,  
'RUNNING')
```

```
        machine_status = 'SHUTDOWN_TRIGGERED';
```

```
        fprintf('EDGE ALERT: Critical vibration (%.2f) detected at  
t=%.2fs! Shutting down motor locally.\n', rms_val, timestamp);
```

```
        % Break the loop to simulate machine stopping
```

```
        cloud_aggregated_data = cloud_aggregated_data(1:w, :);
```

```

        break;
    end

    % Edge to Cloud transfer (Only sending 1 value per window
    instead of 50)
    cloud_aggregated_data(w, :) = [timestamp, rms_val];
end

%% 3. Cloud Architecture (Predictive Analytics)
disp('--- Cloud Tier: Long-term Analytics ---');
% Cloud analyzes the down-sampled RMS trends for predictive
maintenance
time_cloud = cloud_aggregated_data(:,1);
rms_cloud = cloud_aggregated_data(:,2);

% Simple cloud-side predictive trend line (Linear Fit)
if length(time_cloud) > 1
    p = polyfit(time_cloud, rms_cloud, 1);
    trend_line = polyval(p, time_cloud);
    fprintf('CLOUD INSIGHT: Vibration increasing at a rate of %.4f
units/sec.\n', p(1));
end

%% 4. Visualization of the Architecture
figure('Name', 'Industrial Edge Architecture');

subplot(2,1,1);
plot(time, motor_vibration, 'Color', [0.7 0.7 0.7]);
hold on;
yline(vibration_threshold, 'r-', 'LineWidth', 2, 'Label', 'Edge
Shutdown Threshold');
xlabel('Time (s)');
ylabel('Raw Amplitude');
title('Edge Tier: High-Frequency Sensor Data Processing');
grid on;

```

```
subplot(2,1,2);  
plot(time_cloud, rms_cloud, 'bo-', 'LineWidth', 1.5);  
hold on;  
if exist('trend_line', 'var')  
    plot(time_cloud, trend_line, 'g--', 'LineWidth', 2);  
    legend('Data Sent to Cloud (Downsampled)', 'Cloud Predictive  
Trend');  
end  
xlabel('Time (s)');  
ylabel('RMS Vibration');  
title('Cloud Tier: Low-Frequency Aggregated Analytics');  
grid on;
```

Output:

Command Window

```
--- Edge Controller: Real-Time Protection Loop ---  
--- Cloud Tier: Long-term Analytics ---  
CLOUD INSIGHT: Vibration increasing at a rate of 0.0394 units/sec.  
>>
```

Figure 1: Industrial Edge Architecture

